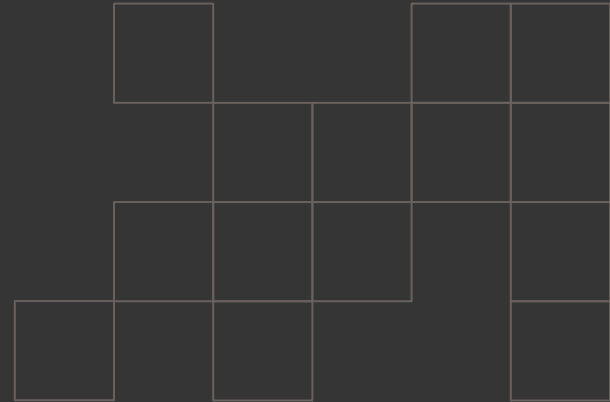


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AI Image Captioning

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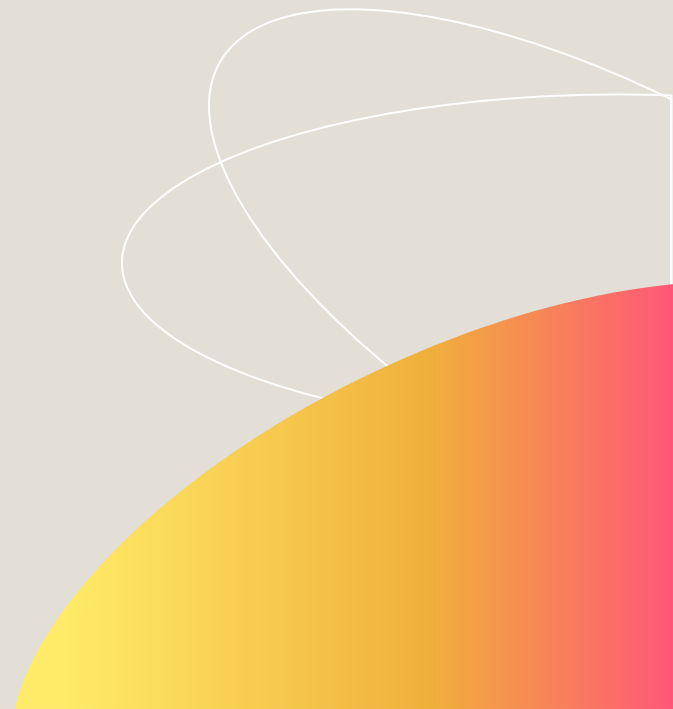
Problem Statement

Problem:

- Images do not have built in descriptions
- Adding captions can improve accessibility for people with disabilities
- Captions also make searching and organizing images easier

Goal:

- Generate captions automatically from images



Approach

- Use a pre-trained BLIP model to generate captions for images and evaluate them.
- Image → Model → Caption

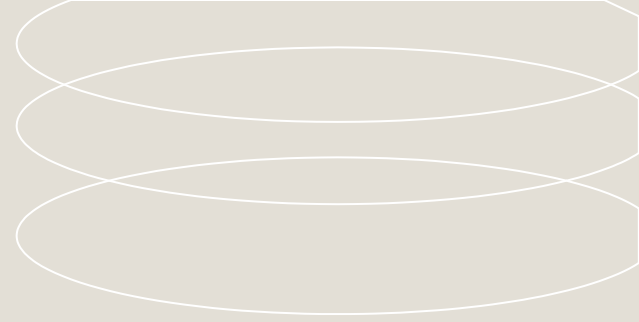
Data

Dataset:

- 15 images
- Categories:
 - Animals
 - People
 - Food
 - Outdoor scenes
 - Objects

Source:

- Unsplash



Results

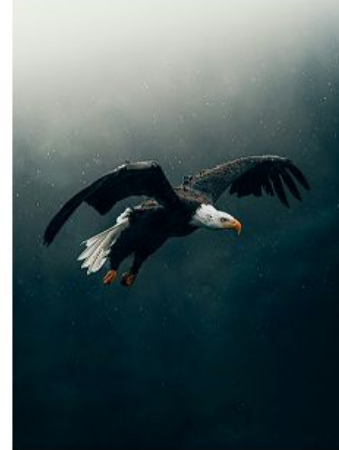
a little girl sitting on a yellow slide



a stack of pancakes with blueberries on a plate



an eagle flying through the air with its wings spread



Evaluation

Metrics:

- Accuracy (1-5)
- Detail (1-5)
- Clarity (1-5)

Accuracy

- 5 = caption matches image very well
- 4 = mostly correct
- 3 = somewhat correct but incomplete
- 2 = weak match
- 1 = incorrect

Detail

- 5 = very descriptive
- 4 = good detail
- 3 = somewhat generic
- 2 = very vague
- 1 = almost no useful detail

Clarity

- 5 = very clear and natural
- 4 = clear
- 3 = understandable but plain
- 2 = awkward
- 1 = unclear

Results:

Accuracy: 4.53

Detail: 4.0

Clarity: 4.67

Evaluation cont.

1	image	caption	accuracy	detail	clarity	notes
2	apple_picking.jpg	a woman and two children in a wagon picking apples from a tree				
3	bird.jpg	a red bird is perched on a branch				
4	burger.jpg	a wooden plate with a hamburger and french fries on it				
5	cooking.jpg	a woman wearing a brown jacket				
6	dog.jpg	a brown dog carrying a stick in the woods				
7	eagle.jpg	an eagle flying through the air with its wings spread				
8	laptop.jpg	a laptop on a wooden table with a cup of coffee				
9	meal.jpg	a white plate topped with meat and vegetables				
10	moon.jpg	a full moon is seen in the dark sky				
11	ocean.jpg	the sky is orange				
12	pancakes.jpg	a stack of pancakes with blueberries on a plate				
13	polar_bear.jpg	a polar bear standing on its hind in the snow				
14	slide.jpg	a little girl sitting on a yellow slide				
15	train_station.jpg	a man wearing a black coat				
16	volcano.jpg	a volcano in the clouds with smoke coming out of it				

1	image	caption	accuracy	detail	clarity	notes
2	apple_picking.jpg	a woman and two children in a wagon picking apples from a tree		5	5	5 Strong caption with clear subject
3	bird.jpg	a red bird is perched on a branch		5	4	5 It could've added the type of bird
4	burger.jpg	a wooden plate with a hamburger and french fries on it		5	5	5 Straight to the point, accurate
5	cooking.jpg	a woman wearing a brown jacket		3	2	3 Left out a lot of details, but was still correct with what it said.
6	dog.jpg	a brown dog carrying a stick in the woods		5	4	5 Strong caption, detailed, very accurate
7	eagle.jpg	an eagle flying through the air with its wings spread		5	5	5 Strong caption, detailed, very accurate
8	laptop.jpg	a laptop on a wooden table with a cup of coffee		5	4	5 Simple and straight forward, could've had a little more detail
9	meal.jpg	a white plate topped with meat and vegetables		5	4	5 Could've specified the kind of meat or vegetables
10	moon.jpg	a full moon is seen in the dark sky		5	5	5 Simple, accurate, straight forward
11	ocean.jpg	the sky is orange		2	2	3 Missed the main subject of the picture, vague
12	pancakes.jpg	a stack of pancakes with blueberries on a plate		5	4	4 Left out other objects, didn't specify blueberries being on the pancakes
13	polar_bear.jpg	a polar bear standing on its hind in the snow		5	5	5 Simple, accurate, straight forward
14	slide.jpg	a little girl sitting on a yellow slide		5	4	5 Could've specified clothes
15	train_station.jpg	a man wearing a black coat		3	3	5 Left out the entire background
16	volcano.jpg	a volcano in the clouds with smoke coming out of it		5	4	5 Could've added a little more detail

Analysis

Strengths:

- Good at identifying main objects
- Clear and readable captions

Weaknesses:

- Often too generic
- Misses fine details
- Struggles with complex scenes

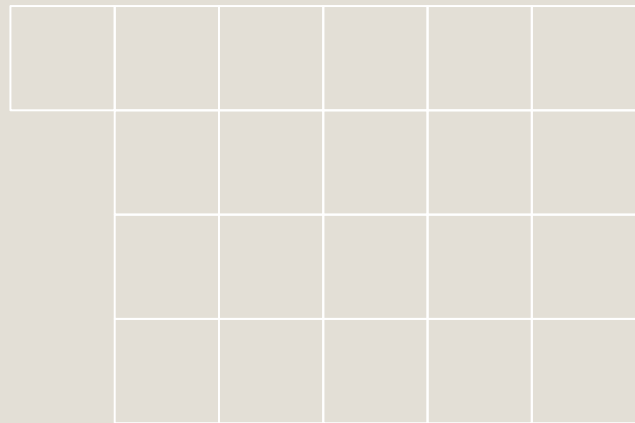
Demo

Live Demo using a smaller sample of 3 images



Limitations:

- Small dataset
- Pre-trained model so no fine-tuning
- Subjective evaluation
- Not very detailed on complex images



Conclusion

- Strong performance on simple images
- Room for improvement on complex scenes
- More training can improve the detailedness
- Detailed was the lowest average score, and lowest overall score

Future Work

- Fine-tune model, or train a new one
- Compare multiple models
- Use automated metrics
- Build user interface



Thank you